BILAL ASIF

Full-Stack Developer | MERN Stack | Node.js | React.js

bilalasif1024@gmail.com | +92-320-799-8854 | Lahore, Punjab, Pakistan

linkedin.com/in/bilal-asif-034272320 | github.com/Bilal-Asif-1

PROFESSIONAL SUMMARY

Full-Stack Developer with a Bachelor of Science in Computer Science and hands-on experience designing and developing scalable web applications using the MERN stack (MongoDB, Express.js, React.js, Node.js). Demonstrated expertise in RESTful API development, JWT-based authentication, role-based access control (RBAC), real-time communication via Socket.IO, and front-end state management with Redux Toolkit. Also experienced with NestJS, TypeScript, and PostgreSQL for backend development. Experienced in Agile/Scrum environments with strong skills in version control (Git/GitHub), code reviews, and iterative production deployments. Adept at both front-end development and back-end architecture, with a focus on responsive web design, database optimization, and secure coding practices.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3

Front-End Development: React.js, Redux Toolkit, Responsive Web Design, Tailwind CSS, Bootstrap, React Hook Form, Yup

Back-End Development: Node.js, Express.js, NestJS, RESTful API Development, Socket.IO, WebSocket, JWT Authentication

Databases: MongoDB, Mongoose, MySQL (indexing, migrations, query optimization), PostgreSQL, Database Design

Tools & Version Control: Git, GitHub, npm, Postman, Visual Studio Code

Methodologies: Agile/Scrum, Software Development Life Cycle (SDLC), MVC Architecture, Code Reviews, Debugging, Technical Documentation

Security: Secure Coding Practices, Role-Based Access Control (RBAC), Input Validation, Data Isolation

PROFESSIONAL EXPERIENCE

Software Engineer (Frontend Developer) | Wimetrix | Lahore, Punjab, Pakistan

January 2023 – Present

Project: Fashion Production Tracking Solution

- Develop and maintain frontend features for a fashion production tracking web application using React.js, contributing to a scalable and responsive user interface.
- Collaborate with backend engineers on MERN stack and NestJS-based services to ensure seamless frontend-backend integration and an optimized user experience.
- Leverage TypeScript and PostgreSQL knowledge to assist in understanding and aligning frontend data models with backend schemas and business logic.
- Contribute to improving system performance, code quality, and overall application reliability through structured development practices and clean coding standards.
- Work closely with cross-functional teams, supporting junior team members and following industry best practices in an Agile/Scrum development environment.

Associate Software Engineer (Internship) | Recurso Labs | Lahore, Punjab, Pakistan

August 2025 – November 2025

Project: KickSpot — Multi-Vendor Sports E-Commerce Marketplace (MERN Stack + TypeScript)

- Designed and developed a full-stack multi-vendor e-commerce web application using TypeScript-based MERN stack, supporting role-based workflows for buyers, sellers, and administrators across a scalable front-end and back-end architecture.
- Implemented JWT-based authentication and Role-Based Access Control (RBAC), eliminating unauthorized access and ensuring complete data isolation across all three user roles.

- Developed and maintained 20+ RESTful APIs using Node.js and Express.js integrated with MySQL; applied indexing strategies and query optimization that reduced average API response time by approximately 35%.
- Built real-time notification system using Socket.IO (WebSocket), enabling live order and inventory updates across all user dashboards with sub-second latency.
- Engineered front-end state management architecture using Redux Toolkit, reducing unnecessary component rerenders and improving UI performance across complex workflows.
- Implemented server-side input validation using Express Validator and front-end schema validation using Yup and React Hook Form, reducing invalid API requests by approximately 40%.
- Participated in system architecture discussions covering ACID properties, CAP theorem, and database design patterns to ensure high reliability and performance at scale.
- Collaborated in Agile/Scrum development environment: contributed to sprint planning, code reviews, technical documentation, and incremental production deployments.

Associate Software Engineer (Internship) | **Punjab Information Technology Board (PITB)** | Lahore, Punjab, Pakistan

November 2024 – February 2025

- Developed CRUD-based web application modules using JavaScript, HTML5, CSS3 following MVC architectural pattern to support internal government administrative workflows.
- Built back-end controllers and views for internal applications, maintaining clean separation of concerns and reusable code structure.
- Implemented form handling, client-side and server-side validation, and database interactions across 5+ internal tools actively used by operational staff.
- Applied debugging techniques and iterative requirement analysis; gained practical exposure to secure coding practices and government-scale application development.
- Collaborated with cross-functional teams to understand large-scale application requirements, contributing to structured software development life cycle (SDLC) processes.

KEY PROJECTS

AgroAI — AI-Powered Smart Farming & Pest Detection System (Final Year Project)

Tech Stack: *MongoDB, Express.js, React.js, Node.js, REST APIs, AI/ML Integration, Real-Time Weather API*

- Engineered a full-stack MERN web application with AI-driven pest detection, delivering automated pesticide recommendations and crop suggestions to farmers based on real-time data.
- Integrated third-party real-time weather data API and designed an inventory management module with multi-role support (farmer, agronomist, administrator).
- Built intuitive, responsive user interface using React.js enabling data-driven agricultural decision-making with measurable improvement in farm productivity.

KickSpot — Multi-Vendor Sports E-Commerce Marketplace

Tech Stack: *React.js, Node.js, Express.js, MySQL, TypeScript, Redux Toolkit, Socket.IO, JWT*

- Delivered end-to-end full-stack development of a production-grade multi-vendor marketplace with real-time features, secure authentication, and optimized database performance.

PetConnect — Pet Adoption & Care Platform

Tech Stack: *React.js, Node.js, Express.js, MongoDB, Tailwind CSS, JWT Authentication*

- Built a full-stack pet adoption web application with features including pet listings, adoption request workflows, image uploads, secure user authentication, and a fully responsive UI.

EDUCATION

Bachelor of Science in Computer Science | Superior University, Lahore, Punjab, Pakistan

September 2021 – September 2025

Final Year Project: AgroAI — AI-Powered Smart Farming & Pest Detection System